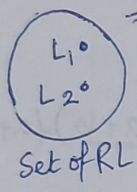
Closure Properties of Regular Language

↳ Closure properties refer to the fact that regular languages are closed under various operations. This means that if you apply these operations to regular languages, the resulting language will also be regular.

Ex: + is closed w.r.t. $\mathbb{Z}$ where $\mathbb{Z}$ is set of integers. ex: $1+2=3$, So $\mathbb{Z}$ is closed under + (addition)
but / is not closed
$3/2 = 1.5$, $1.5 \notin \mathbb{Z}$
∴ the set of integers $\mathbb{Z}$ is not closed under division.

① **Union**: If L_1 and L_2 are regular languages, then union of L_1 and L_2 ($L_1 \cup L_2$) is ~~are~~ also Regular Language

Ex: $L_1 = \{a, b\}$ $L_2 = \{b, c\}$
$L_1 \cup L_2 = \{a, b, c\}$



We pick up 2 RL, such as L_1 and L_2 from set of RL,

Proof: If L_1 is regular language, then there exists regular expression r_1 such that $L_1 = L(r_1)$

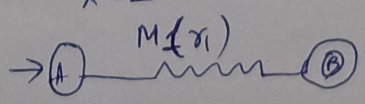
If L_2 is regular language, then there exists R.E. r_2 such that, if r_1 and r_2 are RE, then $L_2 = L(r_2)$

Now, be definition, we know $r_1 + r_2$ are also regular expression that denote the regular language $L(r_1 + r_2) = L(r_1) \cup L(r_2)$

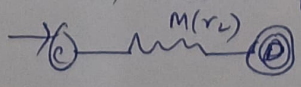
$= L_1 \cup L_2$

(Hence proved)

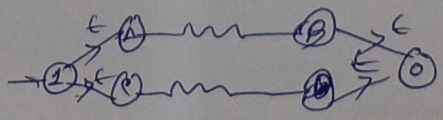
② **Alternate Proof** for L_1 there is FA $M(r_1)$ that accepts $L_1 = L(r_1)$



Similarly for L_2 ... FA $M(r_2)$ that accepts $L_2 = L(r_2)$



Now, union of $M(r_1)$ and $M(r_2)$ also exists, ∴ $L_1 \cup L_2$ is also regular.



② **Concatenation**: $(L_1 \cdot L_2)$ closed under concatenation

If L_1 and L_2 are regular languages then concatenation of L_1 and L_2 $(L_1 \cdot L_2)$ is also regular.

Proof If L_1 and L_2 be two regular languages, then there exists R.E r_1 and r_2 such that $L_1 = L(r_1)$ and $L_2 = L(r_2)$ respectively.

Now, by definition we know if r_1, r_2 are RE then $r_1 \cdot r_2$ is RE that denote the language $L(r_1 \cdot r_2)$ which is regular,

And, $L(r_1 \cdot r_2) = L(r_1) \cdot L(r_2)$

$= L_1 \cdot L_2$.

Ex: ~~$L_1 = \{a\}$, $L_2 = \{b, \epsilon\}$~~

$L_1 = \{a\}$, $L_2 = \{b\}$

$L_1 \cdot L_2 = \{ab\}$ which is regular

Hence proved. (If L_1 and L_2 are RL, then $L_1 \cap L_2$ is also regular language)

③ **Kleen Closure**: If L_1 is RL, then L_1^* is also a RL.

Ex: If $L = \{a\}$, then $L^* = \{\epsilon, a, aa, aaa, \ldots\}$ is also regular.

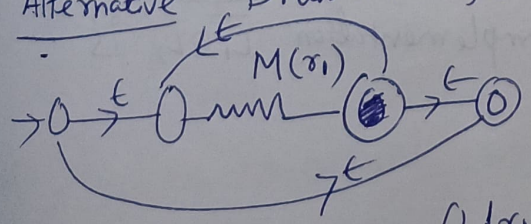
Proof: If L_1 is regular, then there exists, RE, r such that $L_1 = L(r)$

Now, by definition, we know, if r is regular expression, then r^* is also R.E, and for r^*, there is a Regular Language, $L(r^*)$ $= (L(r))^*$ which is equal to $(L(r))^* = (L_1)^*$

$\therefore$ If L_1 is Regular, it implies $(L_1)^*$ is also RL

(Hence proved)

Alternative Draw a FA for $(L_1)^*$ when there is a DFA for L_1



Accepts Language $(L(r_1))^* = (L_1)^*$

④ **Complementation**: If L_1 is regular Language, then the complement of L_1 is also regular.

$$\bar{L_1} = \Sigma^* - L_1$$

Proof: For a given regular Language L_1, there must exist a DFA $M = (Q, \Sigma, \delta, q_0, F)$ that accepts L_1, then the DFA

$\hat{M} = (Q, \Sigma, \delta, q_0, Q-F)$ will accepts $\bar{L_1}$

Complementing DFA means: making final as non final and non final as final states

⑤ **Intersection**: $(L_1 \cap L_2)$ If L_1 and L_2 are RL, then $L_1 \cap L_2$ is also a regular Language.

Ex: If $L_1 = \{a, b\}$, $L_2 = \{b, c\}$, $L_1 \cap L_2 = \{b\}$ which is regular.

Now, according to De Morgan's Law, we can write

$$L_1 \cap L_2 = \overline{\bar{L_1} \cup \bar{L_2}}$$

Now, if L_1 is RL, then by closure under complementation, $\bar{L_1}$ is regular

Now, if L_2 " RL, " " " " " , $\bar{L_2}$ " "

Next, by " " Union, $\bar{L_1} \cup \bar{L_2}$ is regular.

Using closure under complementation $\overline{\bar{L_1} \cup \bar{L_2}}$ is also regular.

(6) Difference: $(L_1 - L_2)$: If L_1 and L_2 be two regular languages, then $L_1 - L_2$ (the difference of L_1 and L_2) is also a regular language.

Ex: $L_1 = \{a, b\}$, $L_2 \{b\}$, then $L_1 - L_2 = \{a\}$ which is regular.

Proof: $L_1 - L_2 = L_1 \cap \overline{L_2}$

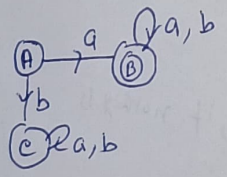
Now, if L_2 is regular, then $\overline{L_2}$ is also regular, because Regular Language is closed under complementation,

Similarly $L_1 \cap \overline{L_2}$ are also regular.

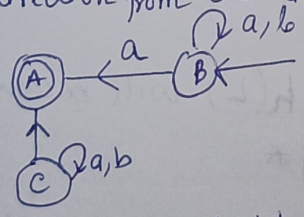
(7) Reversal: If L is a regular language, then L^R (reversal of L) is also a regular language.

Proof: If we have a regular Languge L, then we can construct an NFA, M for that Language. Then we can also construct another NFA M^R that accepts the ~~reverse of the~~ Language L^R (which is the reverse of L). We can construct M^R by using the following step.

Step 1 First draw the NFA for L (for ex: $L = \{$set of all strings start with a$\}$ over $\{a, b\}$

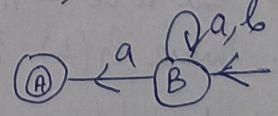
$M =$ 

Step 2: Make final state of M as initial and initial state of M is final and reverse the edge direction from one state to another except the loop.



Step 3 Delete the Unreachable states (from initial states). Here, we delete state C,

$\therefore$ For, L, we have M

$M^R =$ 

and for L^R, we have M^R

here $L^R = \{$set of all strings ends with A$\}$

Def^n of Homomorphism:

Suppose Σ and Γ are alphabets, Then a function

$$h : \Sigma \to \Gamma^*$$

is called a homomorphism. In other words, a homomorphism is a substitution in which a single letter is replaced with a string. The domain of the function h is extended to strings in an obvious fashion if $w = a_1 a_2 \cdots a_n$,

then $h(w) = h(a_1)\, h(a_2)\, h(a_3) \ldots h(a_n)$.

If L is a Language on Σ, then its homomorphic image is defined as $h(L) = \{ h(w) : w \in L \}$ where $L = \{ w : w \in L \}$

Ex: ① Let $\Sigma = \{a, b\}$, and $\Gamma = \{a, b, c\}$ and define h by

$h(a) = ab$,

$h(b) = bbc$

Then $h(aba) = h(a) \cdot h(b) \cdot h(a) = ab \cdot bbc \cdot ab = abbbcab$

If $L = \{aa, aba\}$ is a language, then homomorphic image of L is another language $h(L) = \{abab, abbbcab\}$

Ex: ② $\Sigma = \{a, b\}$, and $\Gamma = \{b, c, d\}$. Define h by

$h(a) = dbcc$

$h(b) = bdc$

Note: homomorphic image of ϵ is always ϵ, if nothing is explicitly given.

If L is the RL denoted by a RE, $r = (a+b)^*)(aa)^*$

then, R.E. for the homomorphic image of L, i.e, for $h(L)$ will become another R.E. $r_1 = (dbcc + (bdc)^*)(dbccdbcc)^*$

③ **Homomorphism:** Let h be a homomorphism. If L is a regular language, then its homomorphic image $h(L)$ is also regular.

Proof: Let L be any RL denoted by some R.E. r. Now, if we wish to find $h(r)$, then we have have to substitute $h(a)$ for each symbol $a \in \Sigma$ of r. So, by definition of R.E, $h(r)$ is also a Regular Expression; hence $h(r)$ obviously ~~denote~~ represents some Regular Language, here the R.L is $h(L)$ $\therefore$ $h(L)$ is Regular

(Hence Proved)

9 Inverse homomorphism : $(h^{-1}(L))$

Ex: $\boxed{\text{Let } h(0) = a, \quad h(1) = b, \quad h(2) = ab}$ homomorphism is defined as

$\Sigma = \{0,1,2\}$, $\Gamma = \{a,b\}$

A language $L = \{abab\}$ is given, we have to find inverse homomorphism $(h^{-1}(L))$ of this language.

Now, we can see that the language L is already in homomorphic form, because its string are in the form of a and b, not in the form of 0, 1, 2, So, we have to convert back it to in the original form

$$h^{-1}(L) = \{\underbrace{0101, 22, 012, 201}_{\text{these are 4 possibilites,}}\}$$

Theorem: If L is a regular language, and h is a homomorphism, then the inverse homomorphism $h^{-1}(L)$ is also regular language.

Proof: If L is regular, then there must be some R.E. r for L, Now, we can get $h^{-1}(L)$ by using h and $h^{-1}(L)$ can be represented by some other R.E. r_1, where every symbol of r_1 can be obtained by $h(\cdot)$.

(Hence proved)